

complement system

补体

CHEN_Zifeng HU_Qi



Jules Bordet 1870-1961

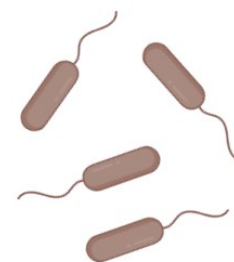
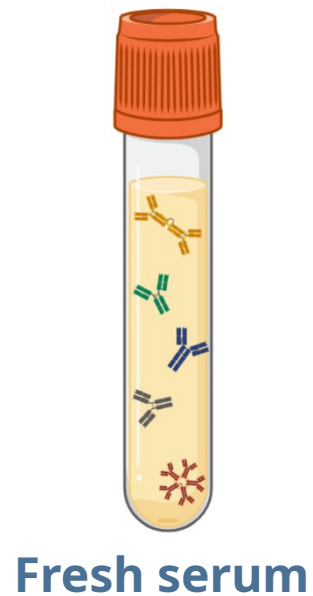
比利时免疫学家

1919 年诺贝尔生理学

或医学奖得主

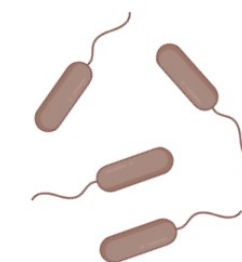
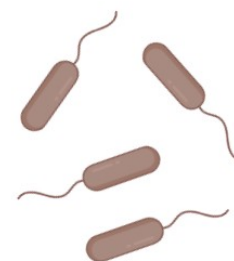
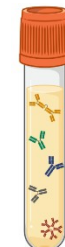
Discover

1890 年



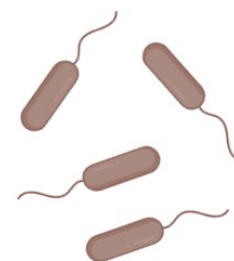
Bacterial lysis

37°C



No lysis

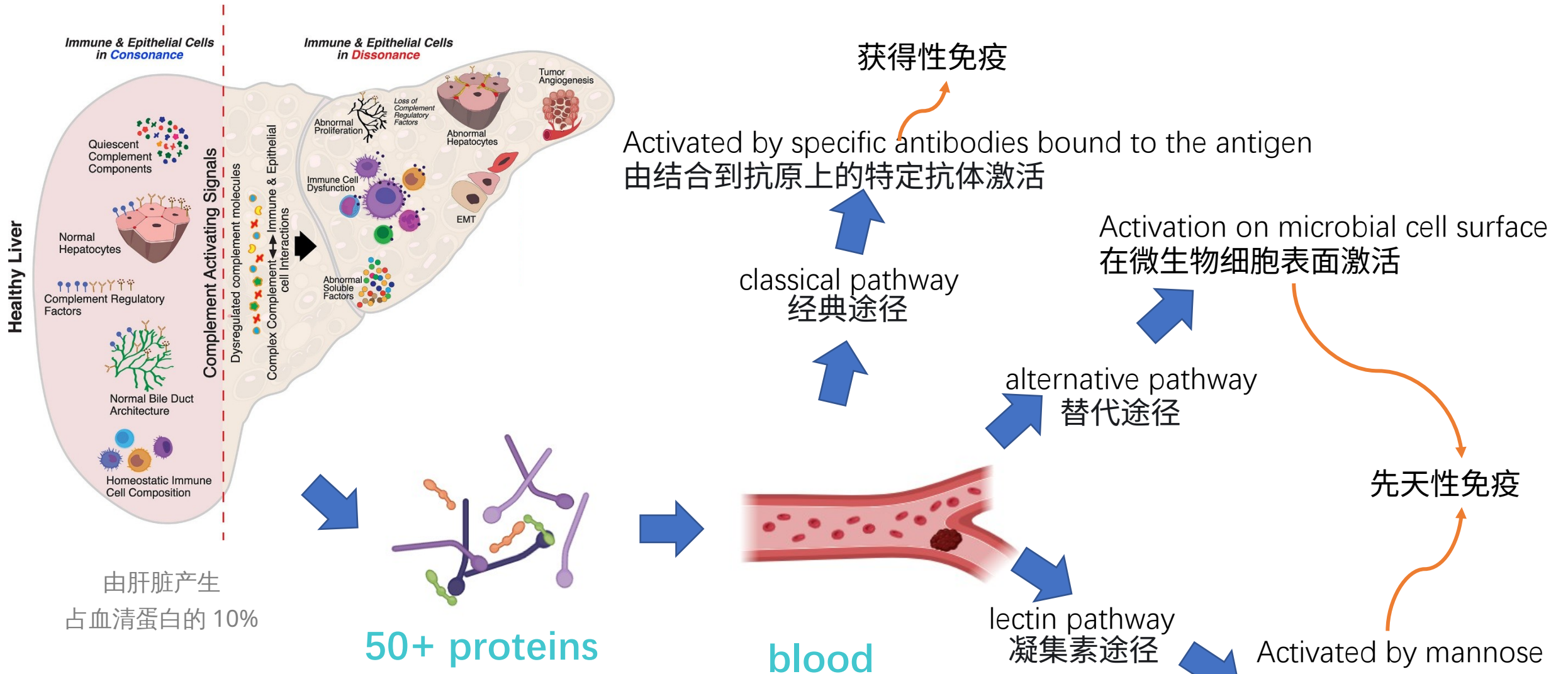
56°C



Bacterial lysis

56°C 37°C

56°C



In the humoral system, complement proteins are inactive or briefly active (several seconds), but they are or are activated after attachment to microorganisms, antibodies, or apoptotic cells.

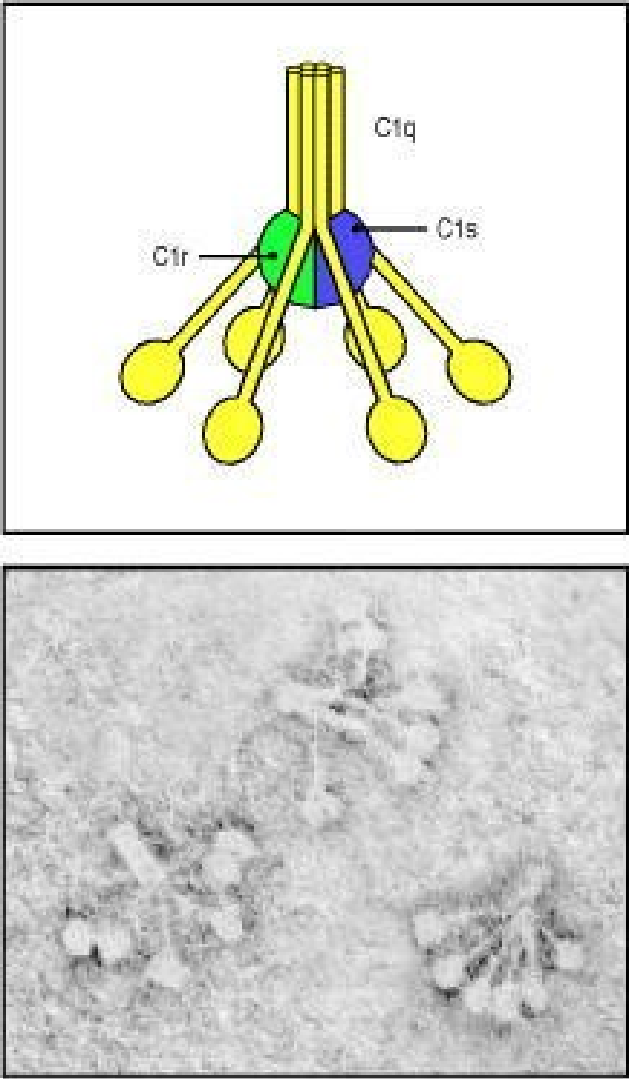
在体液系统中，补体蛋白是不活跃或短暂活跃（几秒钟）的，但它们在附着到微生物、抗体或凋亡细胞后或被激活。

The components of the complement system in the human body

Functional protein classes in the complement system	
Binding to antigen:antibody complexes and pathogen surfaces	C1q
Binding to mannose on bacteria	MBL
Activating enzymes	C1r C1s C2b Bb D MASP-1 MASP-2
Membrane-binding proteins and opsonins	C4b C3b
Peptide mediators of inflammation	C5a C3a C4a

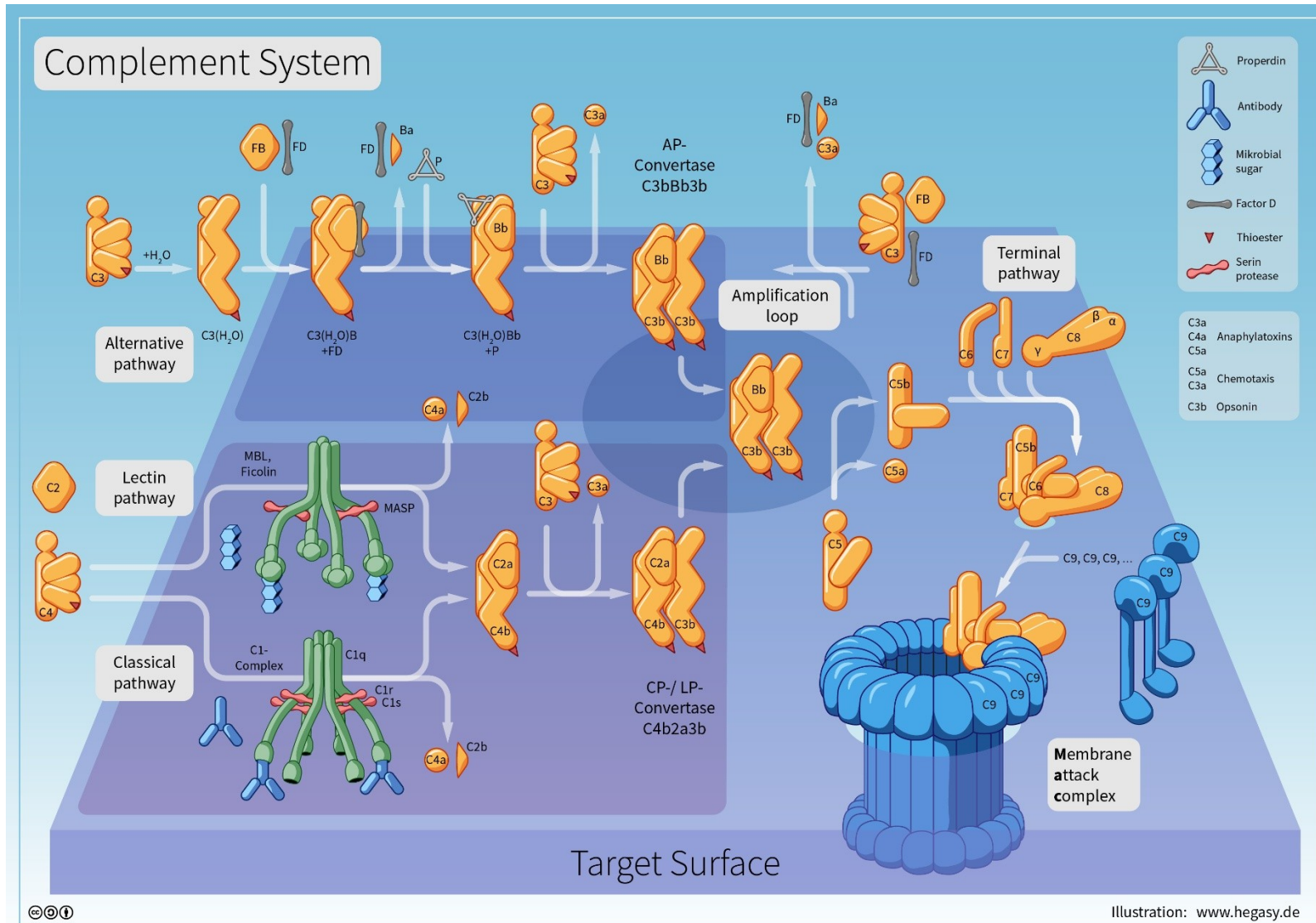
Membrane-attack proteins	C5b C6 C7 C8 C9
Complement receptors	CR1 CR2 CR3 CR4 C1qR
Complement-regulatory proteins	C1INH C4bp CR1 MCP DAF H I P CD59

Functional protein classes in the complement system.
 Janeway CA Jr, Travers P, Walport M, et al. Immunobiology: The Immune System in Health and Disease. 5th edition.
 New York: Garland Science; 2001. The complement system and innate immunity. Available from:
<https://www.ncbi.nlm.nih.gov/books/NBK27100/>



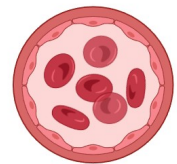
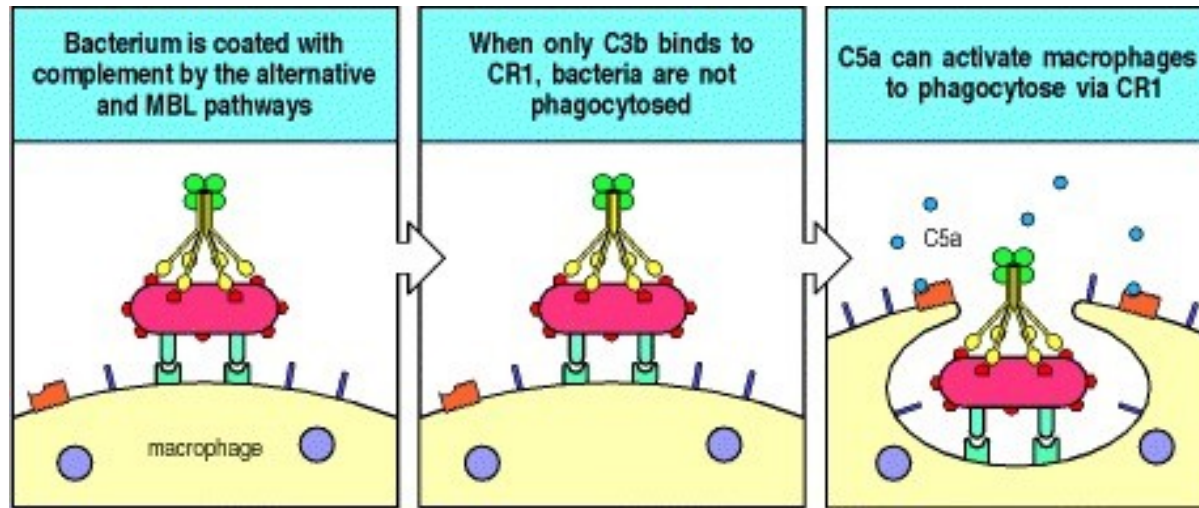
C1 complex

The three activation pathways of the complement system

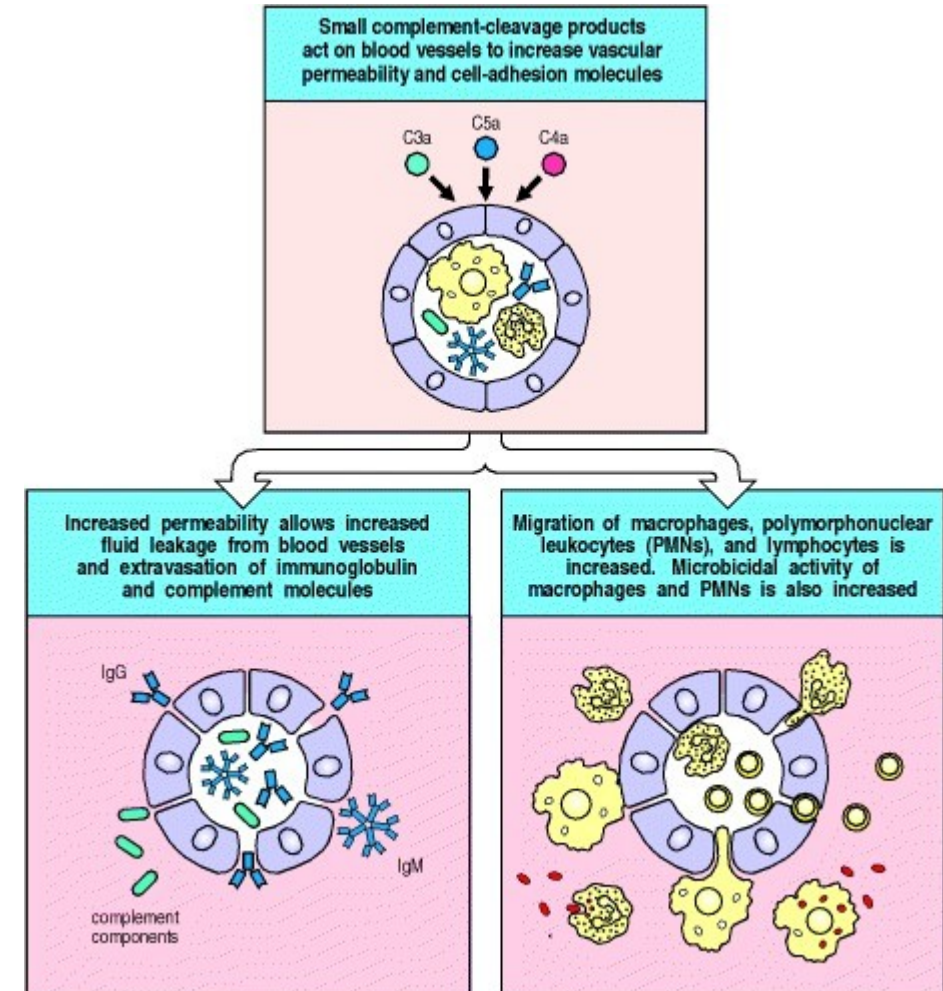
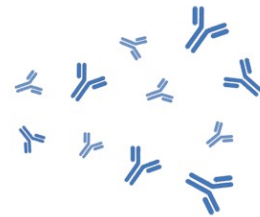


3D structure of membrane attack complex
Thiel S, Petersen SV, Vorup-Jensen T, Matsushita M, Fujita T, Stover CM, Schwaebler WJ, Jensenius JC. Interaction of C1q and mannan-binding lectin (MBL) with C1r, C1s, MBL-associated serine proteases 1 and 2, and the MBL-associated protein Map19. J Immunol. 2000 Jul 15;165(2):878-87. doi: 10.4049/jimmunol.165.2.878. PMID: 10878362.

General functions of complement system



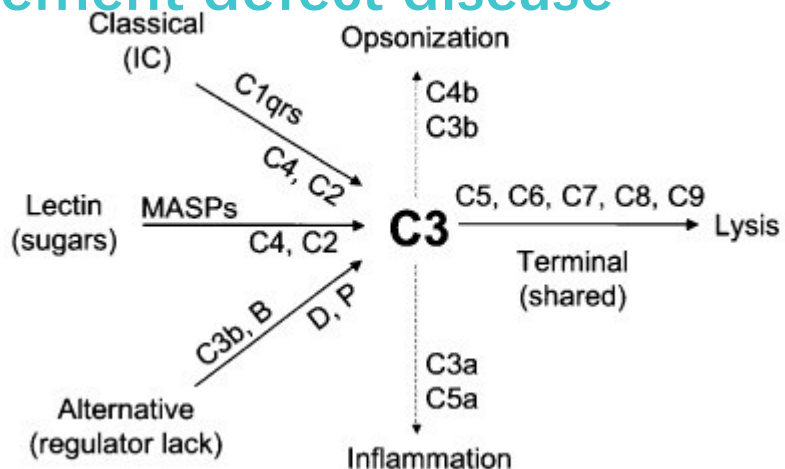
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Janeway CA Jr, Travers P, Walport M, et al. Immunobiology: The Immune System in Health and Disease. 5th edition. New York: Garland Science; 2001. The complement system and innate immunity. Available from: <https://www.ncbi.nlm.nih.gov/books/NBK27100/>

Medical Applications

Complement defect disease



C3 缺陷 → 严重的
反复细菌感染

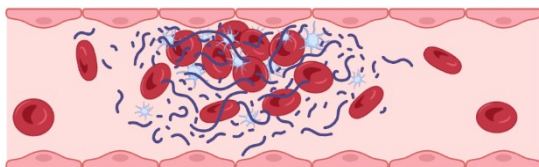
C1 抑制物缺陷 →
遗传性血管性水肿

终末补体 C5-C9 缺陷
→
对奈瑟菌感染高度易感

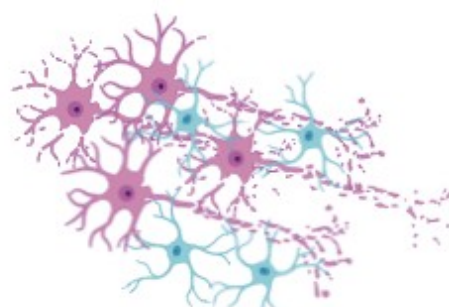
hyperactivation of complement



Autoimmune Diseases
(免疫复合物沉积导致补体消耗)



Abnormal thrombosis
(补体在血管内皮损伤中作用)



Neurodegenerative diseases
(补体介导突触修剪)

Clinical testing and application

- Detect C3 and C4 levels (serological indicators, commonly)
- used for monitoring diseases such as SLE)

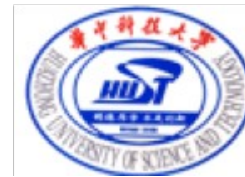
Total complement activity

总补体活性

- CH50 and AH50 functional experiments (detecting the
- activity of classical pathways and alternative pathways)

C3/C4 · CH50 ↓ -- 终末补体途径缺乏
C3 ↓ CH50 ↓ -- 自身免疫性疾病

CH50 ↓ AH50 ↓ -- 终末途径缺陷
CH50 ↓ AH50 · -- 经典途径缺陷
AH50 ↓ CH50 · -- 替代途径缺陷



ANY QUESTION

CHEN_Zifeng HU_Qi

20
25

